

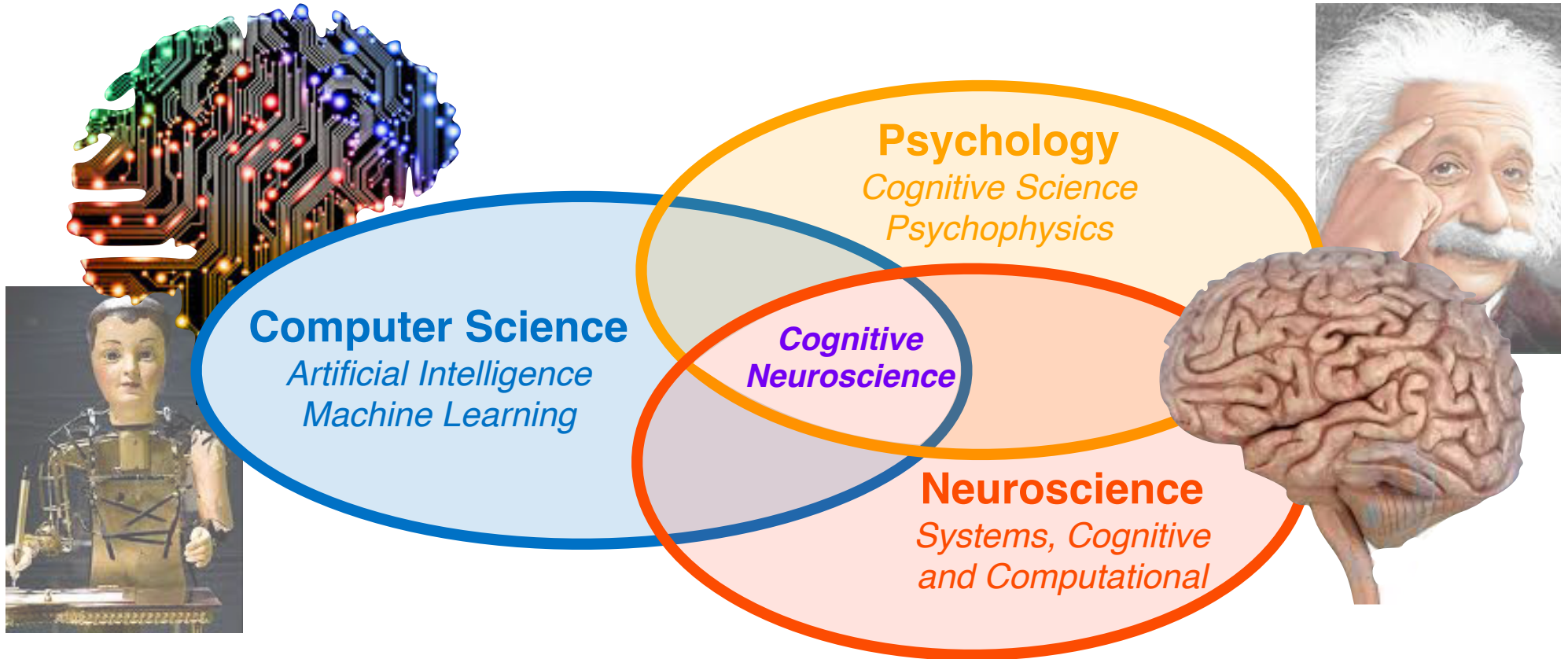
Systems and Cognitive Neuroscience

NEU/PSY/MOL 502A



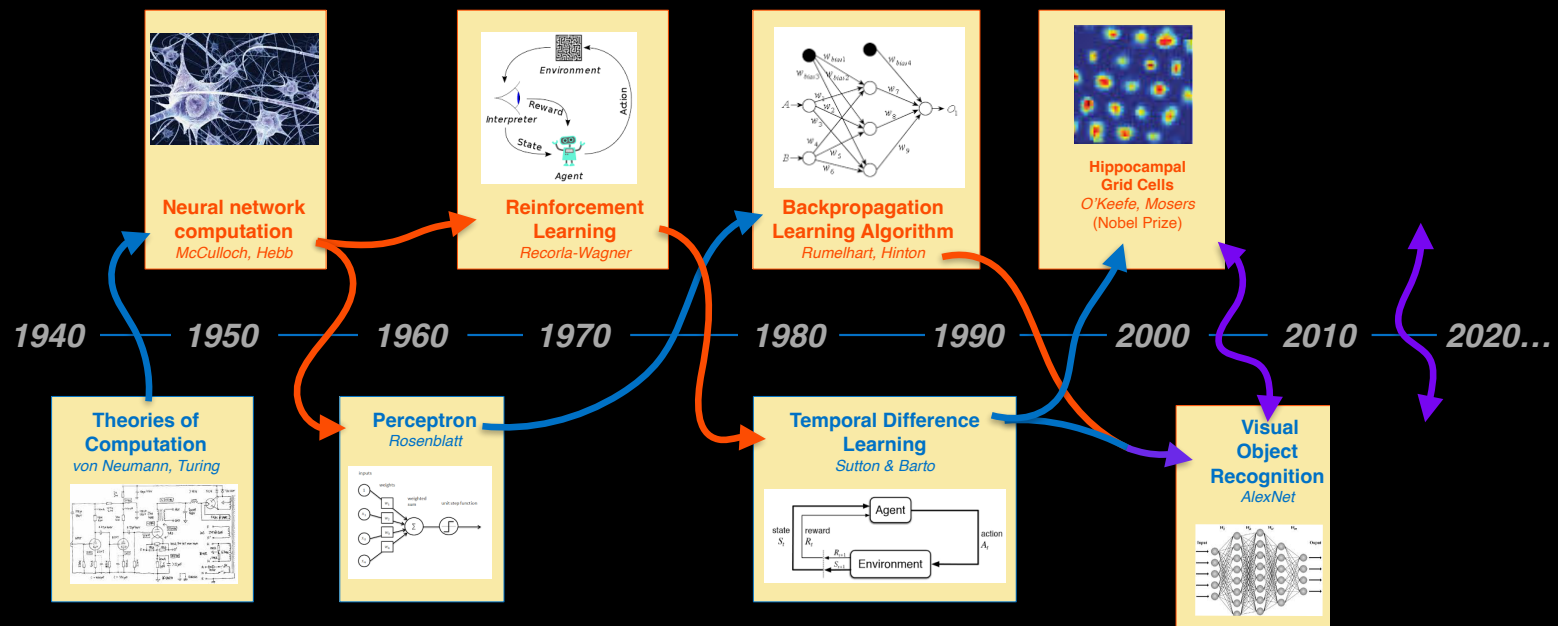
Artificial Intelligence

Natural Intelligence



Long History of Interaction

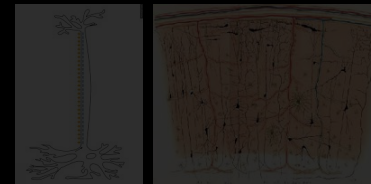
Neuroscience / Psychology



Mathematics / Computer Science

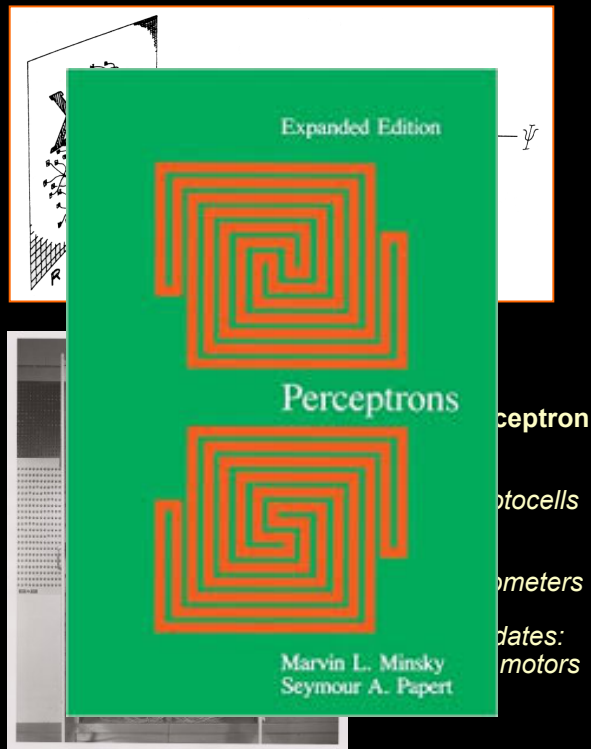
Brief Historical Review

- **Spiking Neurons and Neural Assemblies**
 - McCulloch & Pitts (neuroscience)
 - Hebb (psychology)
 - Norbert Wiener (mathematics)
- **Early neural network models...**



The Perceptron

Frank Rosenblatt, 1957

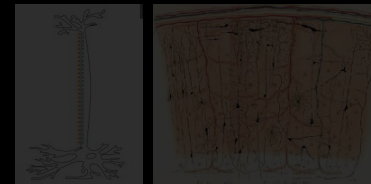


- **Minsky & Papert (1969)**
 - Perceptrons can't learn simple boolean functions (e.g., XOR)
 - ∴ not computationally general

Brief Historical Review

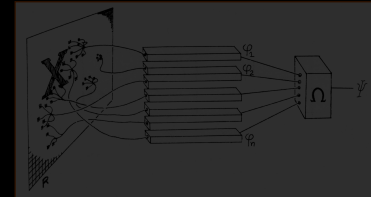
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- Early neural network models

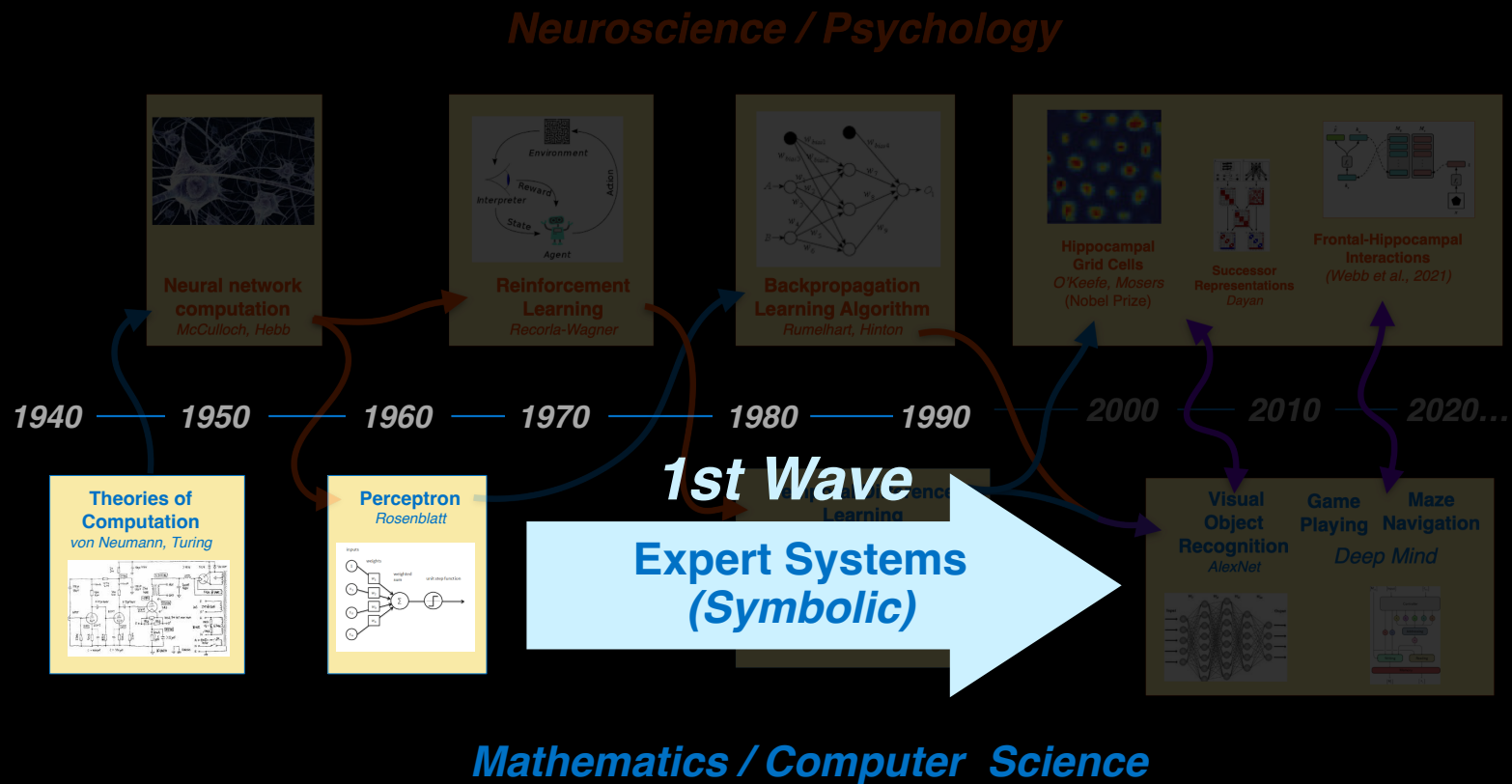
- Rosenblatt's Perceptron
- Minsky & Papert: demise of the Perceptron



- **AI, cognitive science and the symbolic approach:**

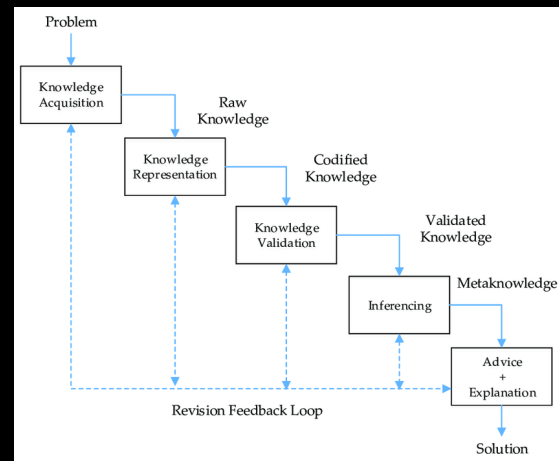
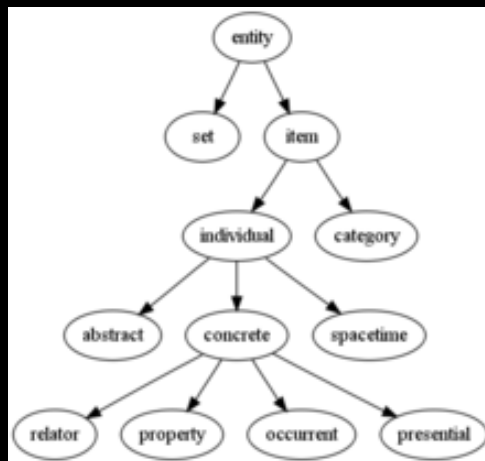
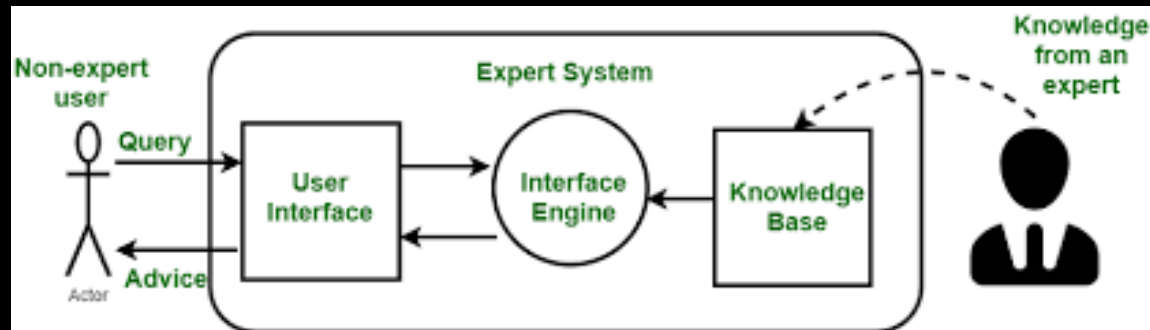
- The physical symbol system hypothesis (Newell & Simon)
- von Neumann Architecture and the computer metaphor
- The golden years of AI...

Brief Historical Review

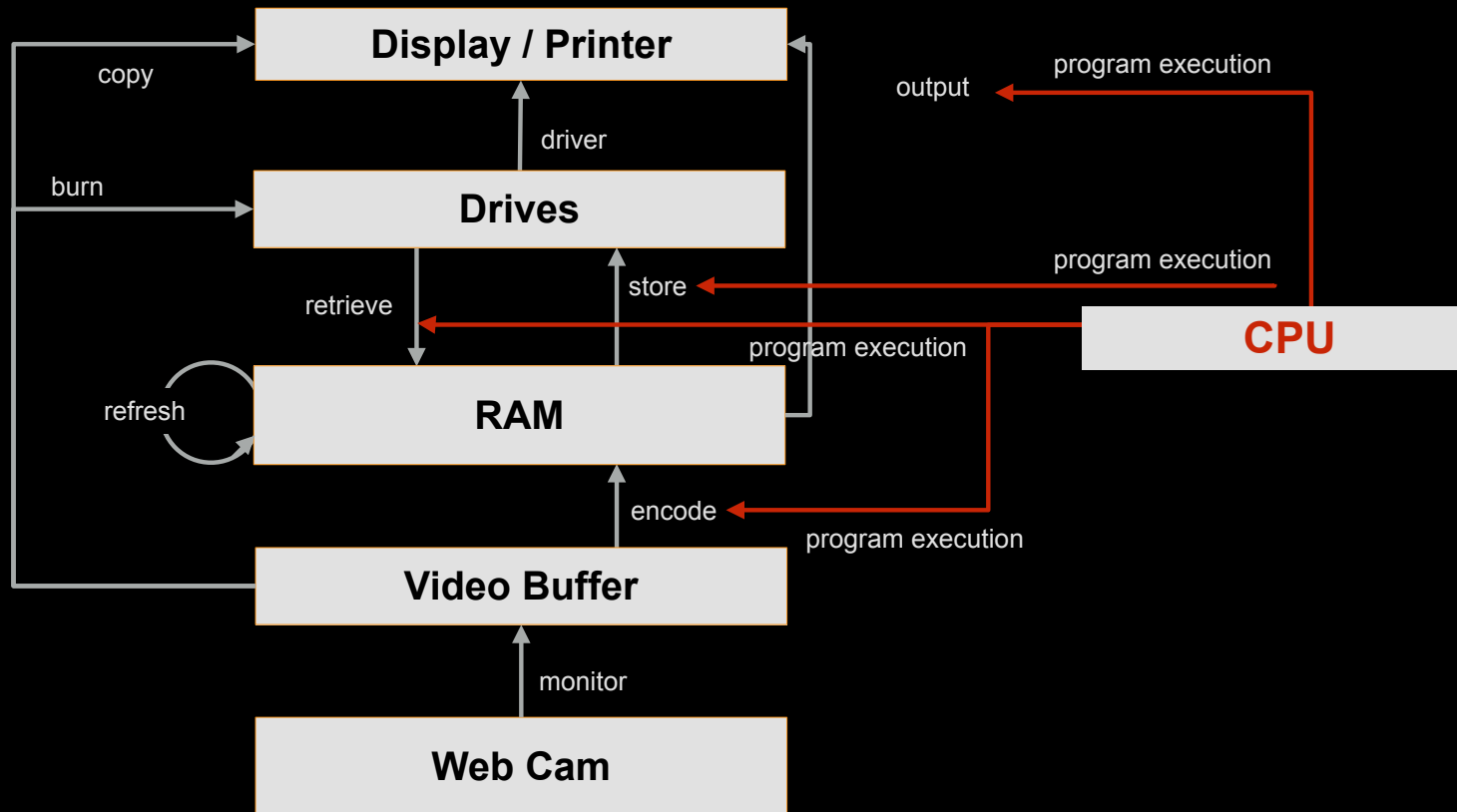


Classical AI

Expert Systems and Knowledge Engineering

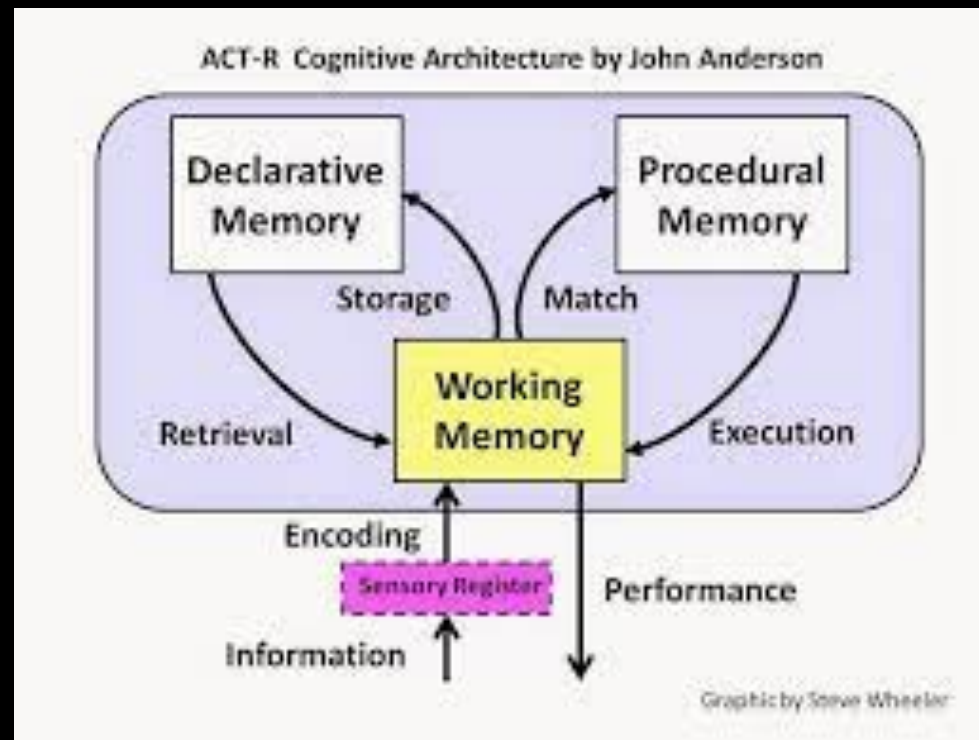


The Von Neumann Computer Metaphor of the Mind



Symbolic Models of Cognition

Production System Models - ACT-R



ACT-R

...the capacity to come up with abstract solutions to problems is one ability that is frequently cited with almost mystical awe. A good example of this is the ability to write recursive programs...

The simulation is capable of solving the same problems as [human] participants. It can actually interact with the same experimental software as the participants, execute the same scanning actions, read the same computer screen, and execute the same motor responses with very similar timing

All that there is to intelligence is the simple accrual and tuning of many small units of knowledge that in total produce complex cognition.

The whole is no more than the sum of its parts, but it has lots of parts

We have studied extensively how people write recursive programs (e.g., Anderson, Farrell, & Sauers, 1984; Pirolli & Anderson, 1985). To test our understanding of the process, we have developed computer simulations that are themselves capable of writing recursive programs in the same way humans do. Underlying this skill are about 500 knowledge units called *production rules*. For instance, one of these production rules for programming recursion, which might apply in the midst of the problem solving, is

IF the goal is to identify the recursive relationship in a function with a number argument

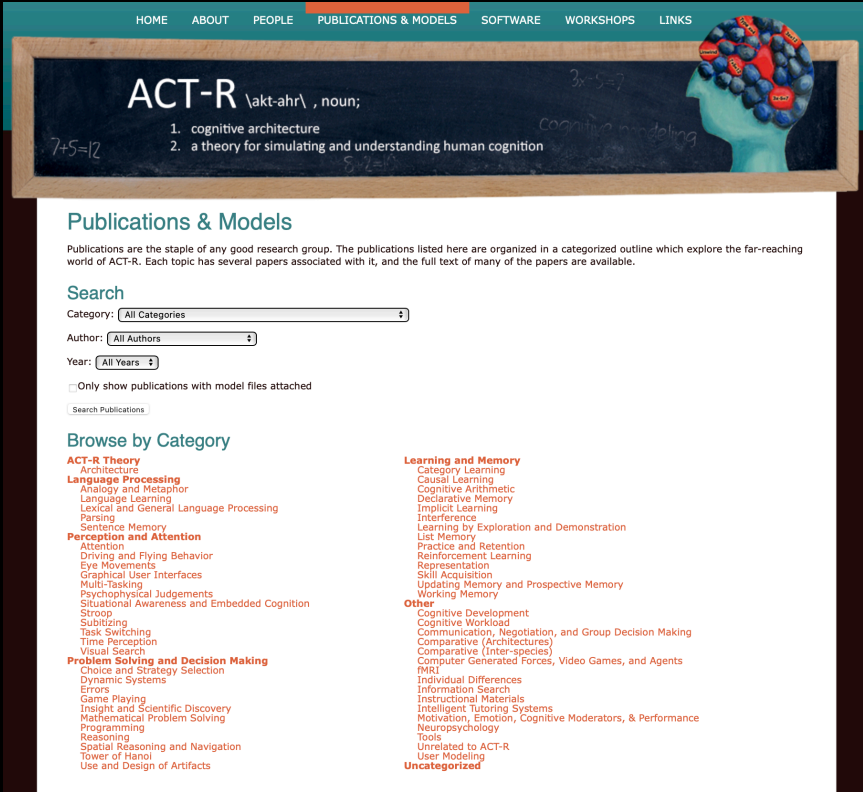
THEN set as subgoals to

1. Find the value of the function for some N
2. Find the value of the function for $N-1$
3. Try to identify the relationship between the two answers.

Thus, in the case above, this might lead to finding that $\text{factorial}(5) = 120$ (Step 1), $\text{factorial}(4) = 24$ (Step 2), and that $\text{factorial}(N) = \text{factorial}(N-1) \times N$ (Step 3).

We (e.g., Anderson, Boyle, Corbett, & Lewis, 1990; Anderson, Corbett, Koedinger, & Pelletier, 1995; Anderson & Reiser, 1985) have created computer-based instructional systems, called *intelligent tutors*, for teaching cognitive skills based on this kind of production-rule analysis. By basing instruction on such rules, we have been able to increase students' rate of learning by a factor of 3.

ACT-R

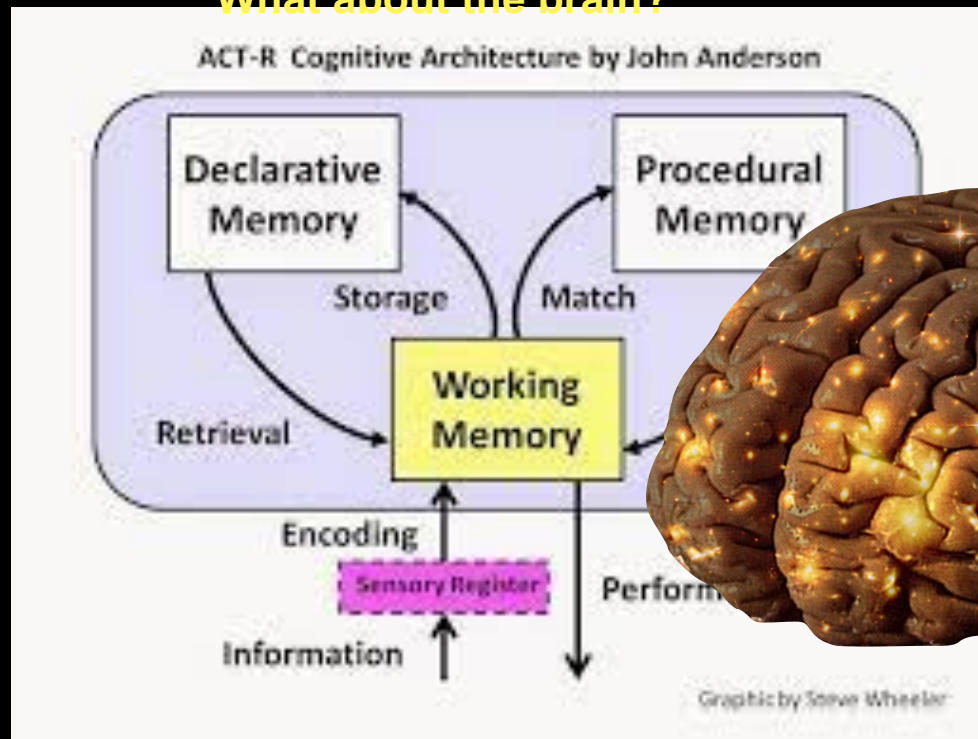


ACT-R

The whole is no more than the sum of its parts, but it has lots of parts

Really... lots of parts

What about the brain?



Brief Historical Review

- **Roots**

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- Hebb (psychology)

- **Early neural network models**

- Rosenblatt's Perceptron
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- **AI and cognitive science**

- The physical symbol system hypothesis (Newell & Simon)
- von Neumann Architecture and the computer metaphor
- Knowledge engineering and the golden years of AI
- Production system models of cognitive function

- **Limits of the symbolic approach**

- Knowledge engineering (expert systems): programming vs. learning
- Combinatorial explosion in highly contextual domains
face recognition, natural language processing...

RED

Three white cat silhouettes are positioned behind the letters of the word 'RED'. The first silhouette is behind the 'R', the second is behind the 'E', and the third is behind the 'D'. They are arranged in a slightly overlapping manner, with the 'E' silhouette being the most prominent.

THE

CAT

PFB



THE

CAT

Brief Historical Review

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- **Limits of the symbolic approach**
 - Knowledge engineering: programming vs. learning
 - Combinatorial explosion in highly contextual domains
face recognition, natural language processing...
 - 100 step challenge: the **brain** is instructive here, but...

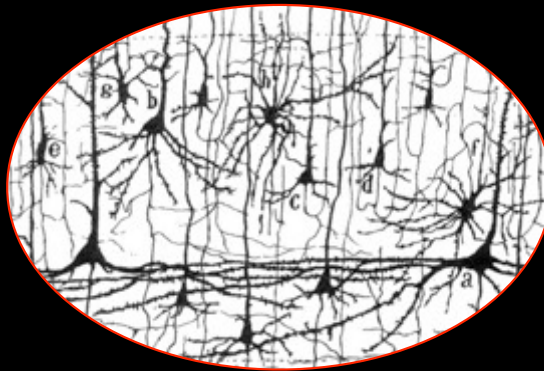
Scale of the Problem

100 billion neurons

100 thousand connections/neuron

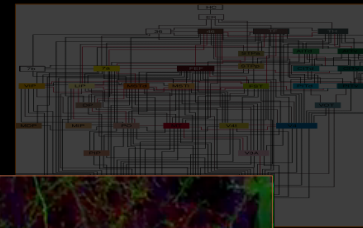
= 100 trillion connections

More potential circuits than molecules in the universe

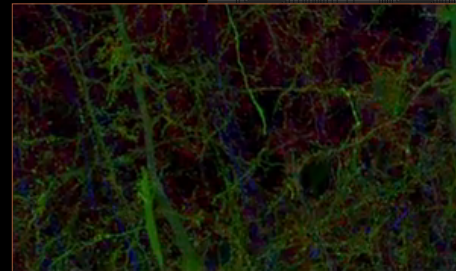


“Bottom-Up” Approach

Go for entire wiring diagram...



“Connectomics”
(map every connection)
but...



The mouse brain alone
will take...

???

And we’d only know structure, not *function*

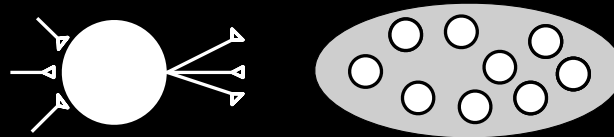
The Connectionist (PDP) Approach

- Brain-*like* computational *architecture*
 - biologically-inspired/plausible processing mechanisms
- Distributed representation
 - statistical structure / graded processing
- Parallel processing
 - can meet 100 step challenge
- Self-organization through experience
 - general purpose statistical learning algorithms

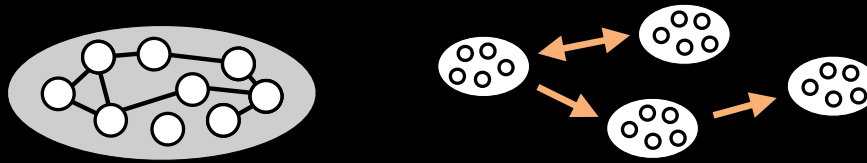
Skip PDP intro

Basic Elements

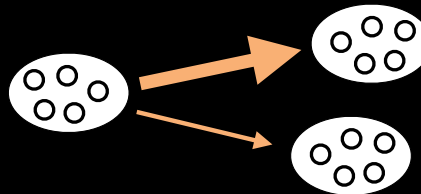
- **Units / Modules** ($\approx$ neuron or population of neurons)



- **Connections / Pathways** ($\approx$ synapses / projections / circuits)

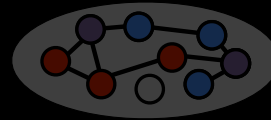


- **Learning Rules** ($\approx$ synaptic plasticity)

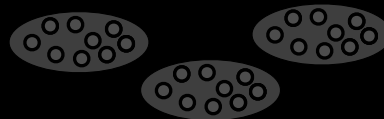


Psychological Constructs

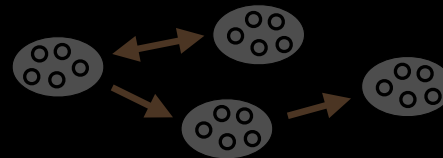
- **Representation** (units)



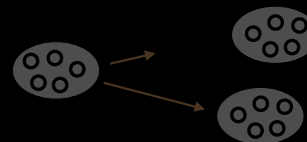
- **Functions** (modules)



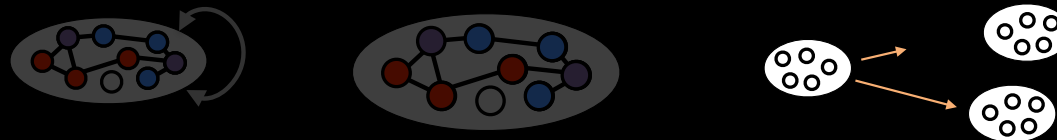
- **Processing** (flow of activity)



- **Learning** (weight modification)



- **Memory** (active maintenance, pattern completion, or weight modification)



Representation

- **Units** ($\approx$ neurons or population of neurons)

- **Activity level**

- ($\approx$ firing frequency or probability of firing)

- **Activation (transfer) function**

- Integrate & fire
 - Thresholded (piecewise) linear
 - Continuous valued (sigmoid function, e.g. logistic)

- **Noise**

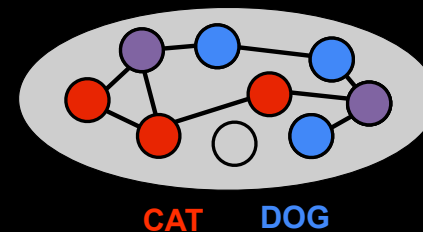
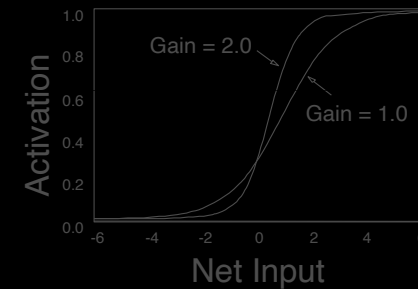
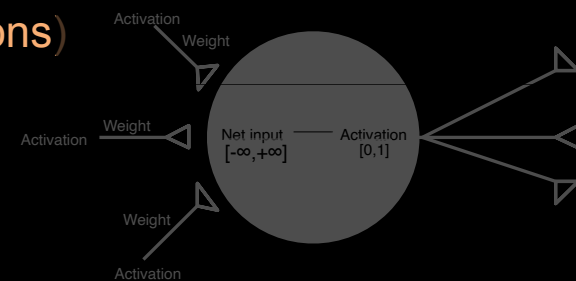
- **Modulation**

- **Patterns of activity** ($\approx$ population code)

- **Distributed representation**

- **Relationships:**

- associations between units/patterns
 - overlap of patterns



Specialization

- **Modules** ($\approx$ brain areas)

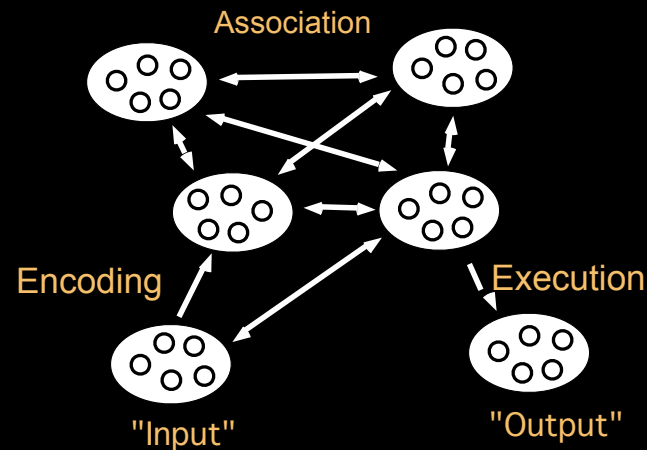
sets of units responsible for:

representing a particular type of information

stimulus (input), semantic (hidden), motor (output) etc.

carrying out a particular function

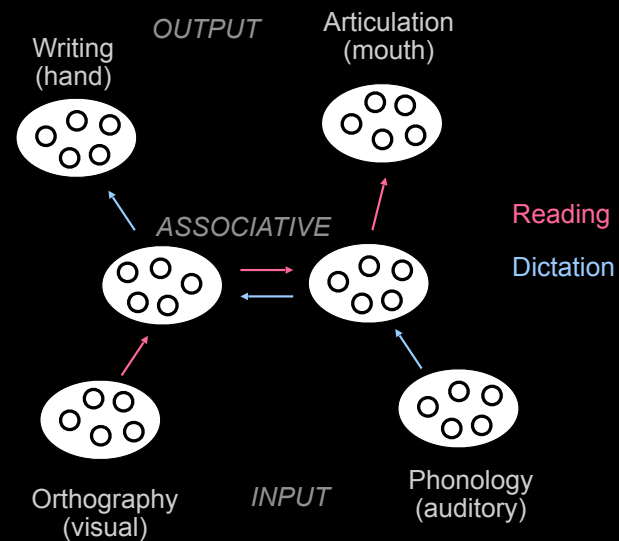
sensory encoding (input), associative (hidden), motor control (output), etc.



Processing

- Flow of activity among units / between modules

- Input output mappings (pathways)
- Interference
- Control = modulation

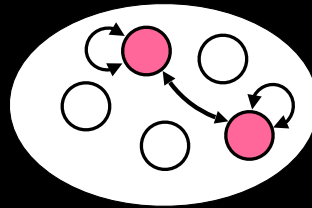


Learning

- **Weight modification** ($\approx$ synaptic plasticity)
 - **Unsupervised** (self-organizing)
 - **Simple associative** (Hebbian)
 - **Competitive** (K-winner take all)
 - **Supervised (trained)**
 - **Reinforcement** (value prediction — scalar)
 - **Structured** (object / event prediction - vectorial)

Memory

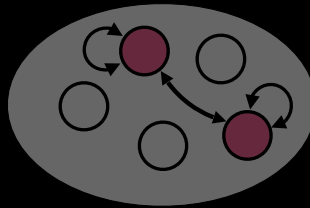
- **Short Term**
 - sustained pattern of activity



Memory

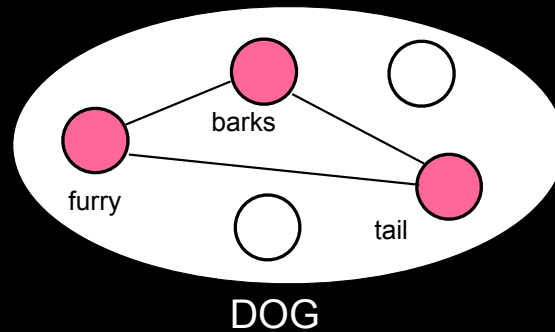
- **Short Term**

- sustained pattern of activity



- **Long Term**

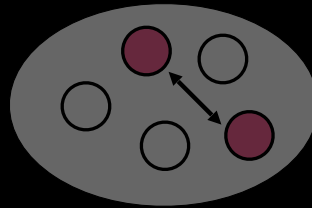
- changes in connections among associated features



Memory

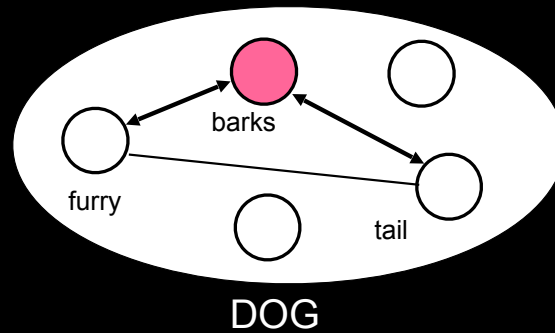
- Short Term

- sustained pattern of activity



- Long Term

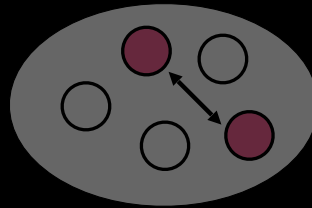
- retrieval:



Memory

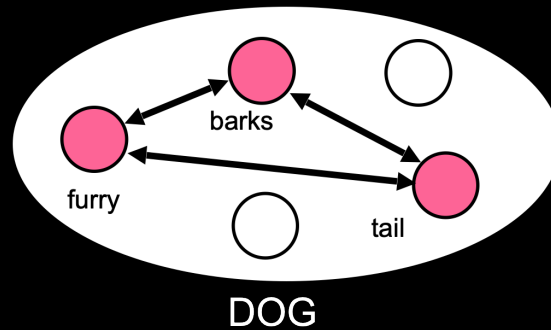
- **Short Term**

- sustained pattern of activity



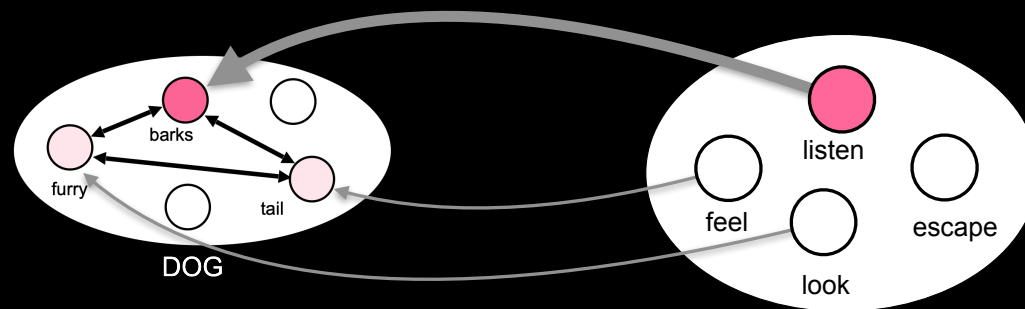
- **Long Term**

- retrieval: reactivation of a whole pattern from its parts



Control

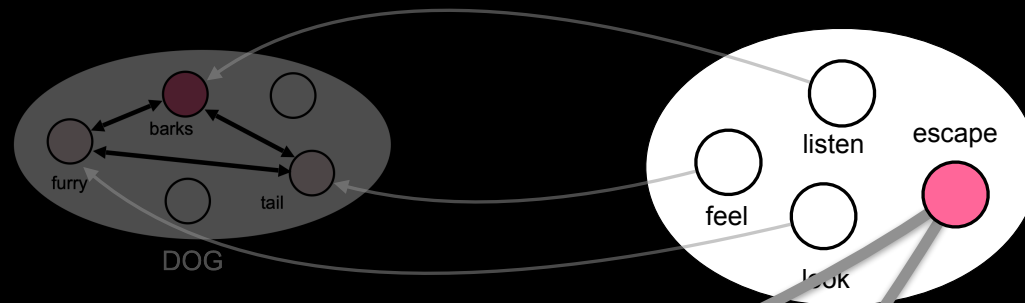
- **Attention**
 - selection of some **features to activate**



Control

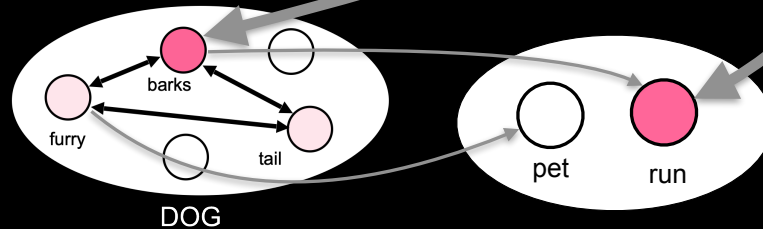
- **Attention**

- selection of features to activate



- **Execution**

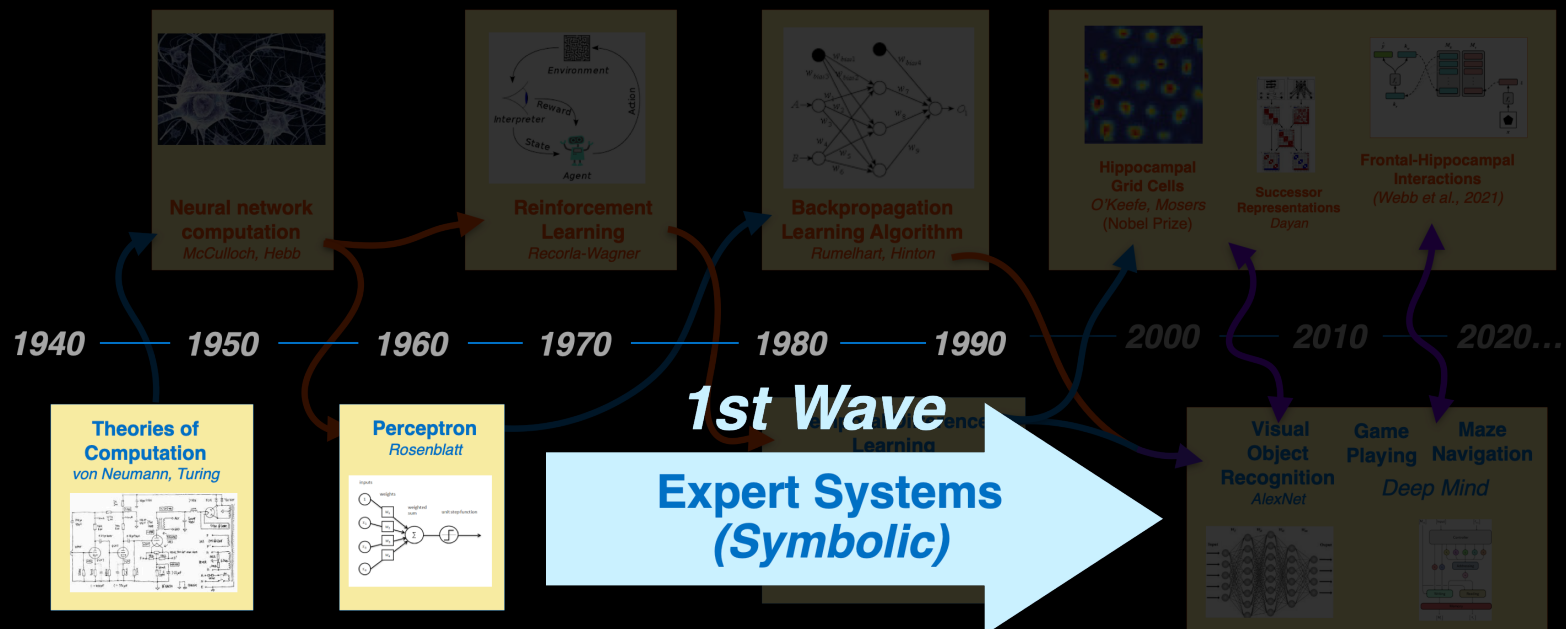
- selection of pathways for
flow of activity



Early Connectionist Models

- Good at doing what the brain does easily
(and what traditional computers do poorly):
lots of fine grained, simultaneous (parallel) mutual interactions among features:
 - visual pattern recognition
 - language processing
 - generalization / pattern completion
- Bad at doing what the brain does poorly
(and what traditional computer programs do easily):
regularly structured, highly general, abstract processes
 - arithmetic
 - reasoning
 - planning
- However, things have changed...

Neuroscience / Psychology



Mathematics / Computer Science

“Deep Learning”



Dave Rumelhart
(UCSD / Stanford)



Yann LeCun
(NYU)



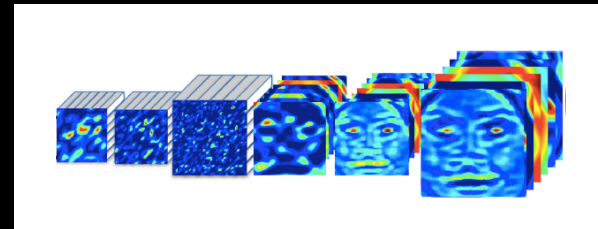
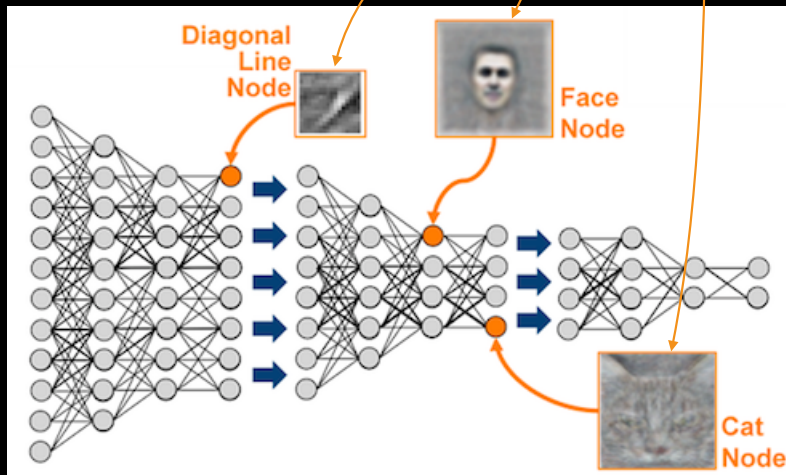
Geoff Hinton
(Toronto / Google)



Bruno Olshausen
(Redwood Institute)



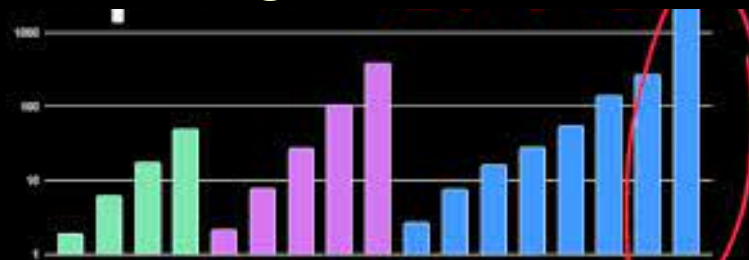
Fei Fei Li
(Princeton/Stanford)



“Deep Learning”



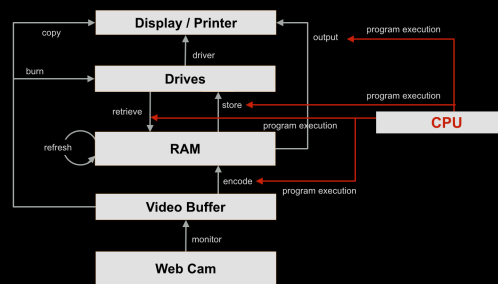
Google **GEMINI**



1.5 TRILLION Parameters

Extremes of a Continuum

Symbolic



logical

serial

discrete

localized

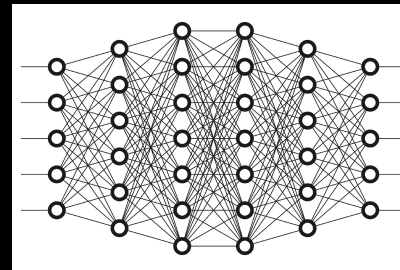
episodic

variance

symbol manipulation

flexible

Connectionist



statistical

parallel

continuous

distributed

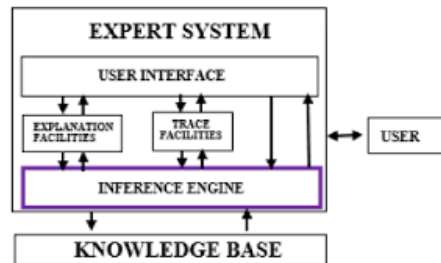
semantic

bias

function approximation

efficient

Artificial Intelligence



Symbolic

Knowledge:

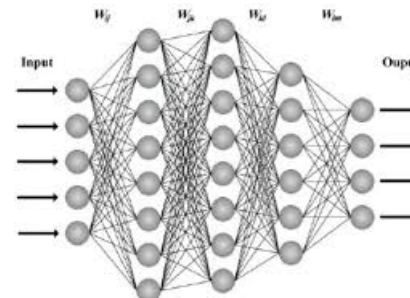
explicitly represented
expressions and procedures

- ✓ *explanation*
- ✗ *domain specific*

Configuration:

programming

- ✓ *flexible*
- ✗ *hand-coded*



Connectionist

implicitly represented
connection weights

- ✓ *efficient*
- ✗ *domain specific*

learning

- ✓ *learns from experience,*
- ✗ *but only when trained*

Clash of the Titans



Complementary Approaches

- **Symbolic approach**

- compositional, **context-free**:

2 + 2

47 + 2

$n + m \leftarrow$ the interpretation of m is not affected by n

- **PDP / connectionist approach**

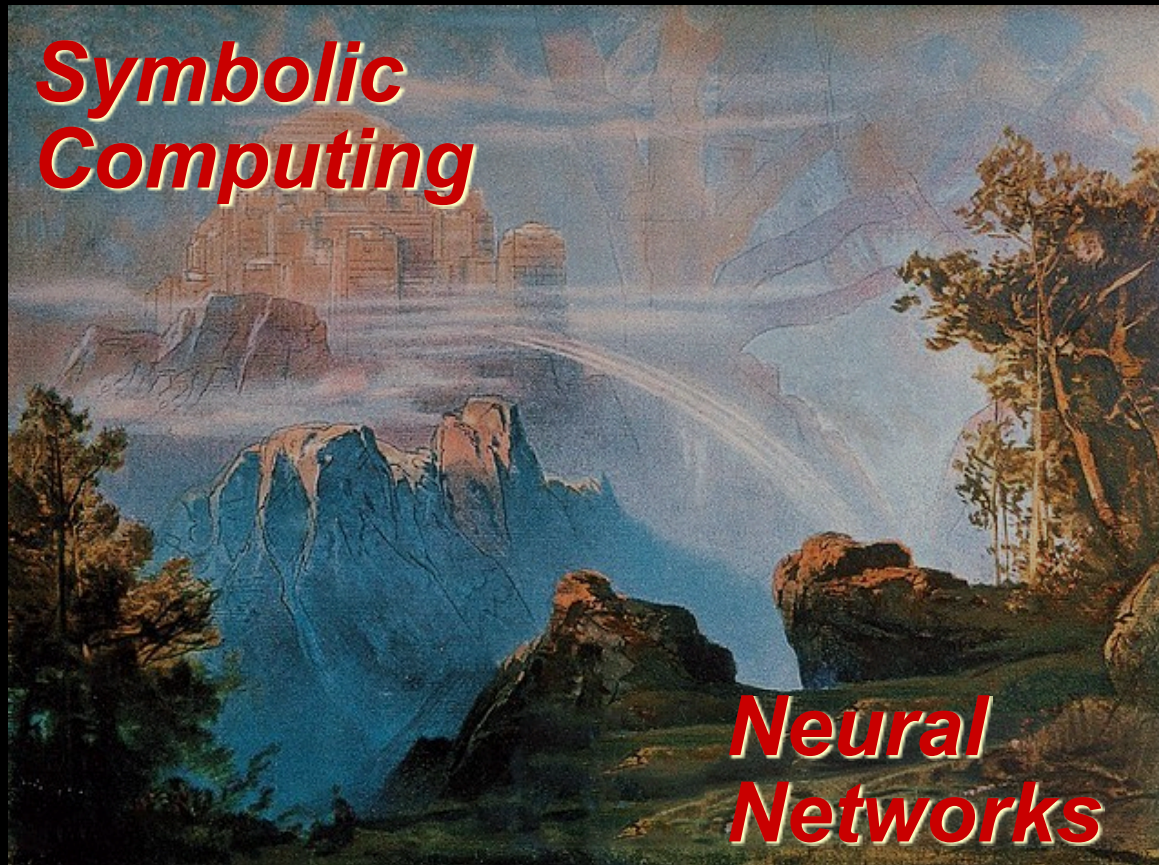
- **context-sensitive**:

river + bank

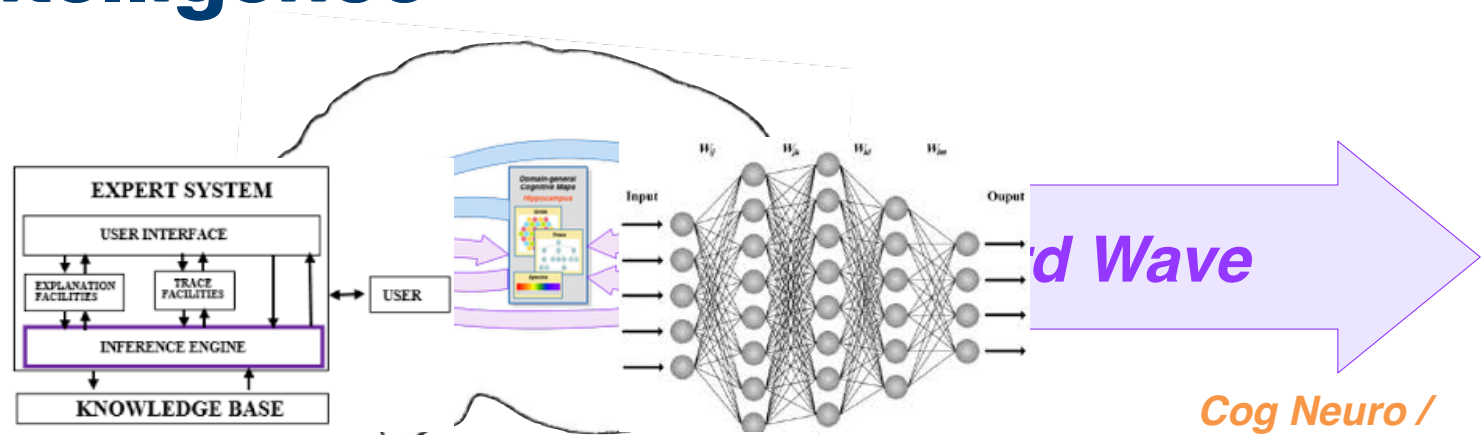
savings + bank

$n + m \leftarrow$ the interpretation of m depends on n

Shangri-La



Artificial Intelligence



Symbolic

Knowledge:

explicitly represented
expressions and procedures
✓ explanation
✗ domain specific

Configuration:

programming
✓ flexible
✗ hand-coded

Connectionist

implicitly represented
connection weights
✓ efficient
✗ domain specific

learning

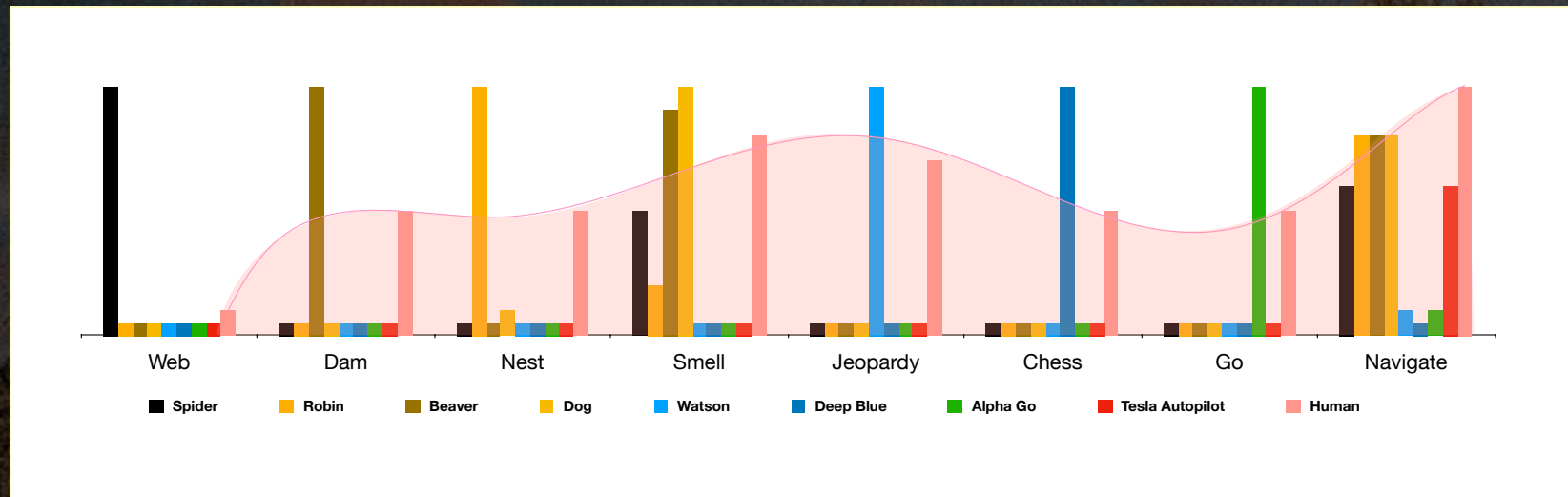
✓ learns from experience,
✗ but only when trained

**Cog Neuro /
System Level
Brain Modeling**

**efficient
generalization**

**autonomous
adaptation**

Human Brain = Existence Proof



“Sweet spot” between flexibility and efficiency

- Near limitless range of tasks at adequate performance - flexibility
- With *reasonable* amounts, and often little or *no* training - sample efficiency
- ~20 watts, often with parallel performance - processing efficiency

How does it accomplish this?

Shangri-La?

- **Challenge:**

- Integrate flexibility of *symbolic processing* in traditional architectures
- with efficiency of *function approximation* in neural networks

- **Current efforts:**

- **Neuro-symbolic** approaches:

- ♦ start with pre-specified symbolic primitives (“core knowledge”)
- ♦ use deep learning to combine these (e.g., “program induction”)

- **“Neo-connectionist”** approaches:

- ♦ use deep learning for “end-to-end” training of neural networks
- ♦ inductive biases that favor **abstraction**
 - *training*: curricular learning, meta learning
 - *architecture & processing*: attention, external memory

- **Still not there...**

- ♦ How does symbolic computing and the capacity for intelligence arise in the brain?

Building Bridges

1980-2000

Neurobiology



Psychology

Building Bridges

2015 →

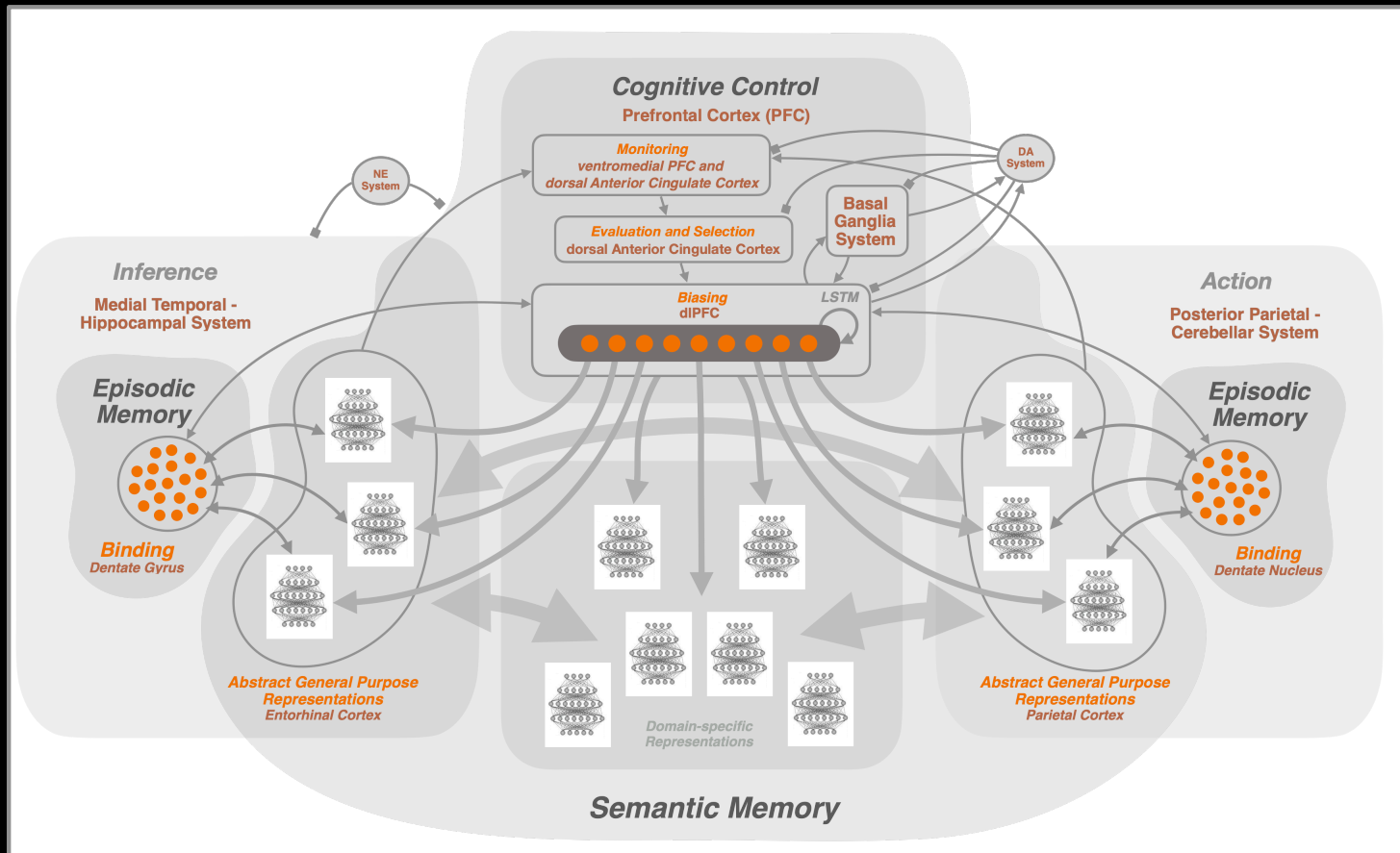
Psychology

Neurobiology



Computer
Science

Where we'd like to be...



Levels of Analysis (Marr)

- **Computational** (*cognitive science*)

- What is the overall goal?

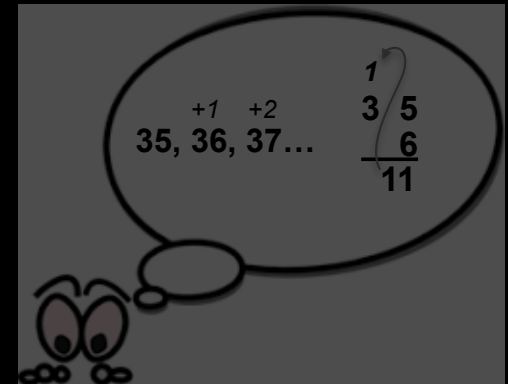
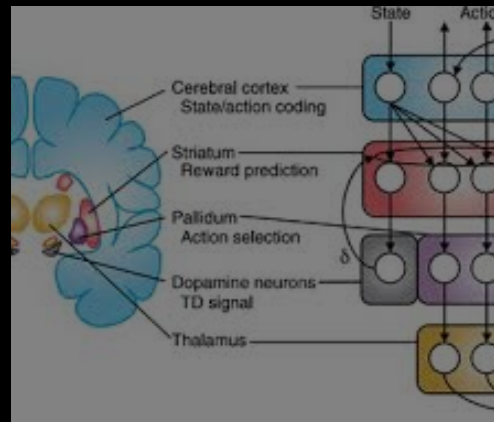
$$35 + 6 = ?$$

- **Algorithmic** (*cognitive psychology / cognitive neuroscience*)

- What strategy is used?

- **Implementational** (*neuroscience*)

- How is it physically realized?



Levels of Analysis (in reality)



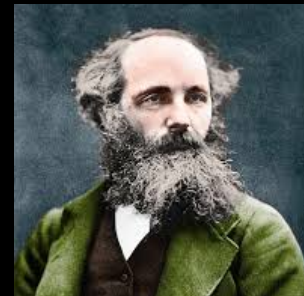
- Behavior of a coin:
 - Trajectory?



Newton

$$\begin{aligned}v &= u + at \\v^2 &= u^2 + 2as \\s &= ut + \frac{1}{2}at^2\end{aligned}$$

or



Maxwell

?

Levels of Analysis (in reality)

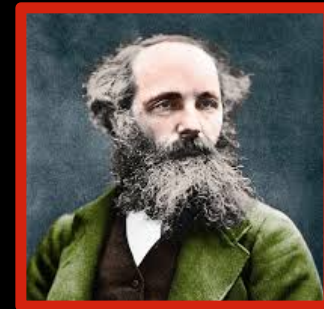


- Behavior of a coin:
 - Trajectory?
 - Melting point?



Newton

or



Maxwell

?

$$\Gamma_{ij} = \left(\delta_{ij} + x_i \frac{\partial \ln \gamma_i}{\partial x_j} \right)$$

Levels of Analysis (in reality)

- “More is different” (*Anderson, 1972*)
- Emergent properties
- Instrumentalism:
 - what is the observational variance you wish to capture?
 - what is the best way to capture that?

Levels of Organization

Physical

- **Organism**
 - Animal
- **Organ**
 - Brain
- **Components**
 - Lobes
- **Maps & Zones**
 - Areas, layers & columns
- **Cells**
 - Neurons
- **Organelles**
 - Membranes & Synapses
- **Molecules**
 - Transmitters & Receptors

Functional

Behavior

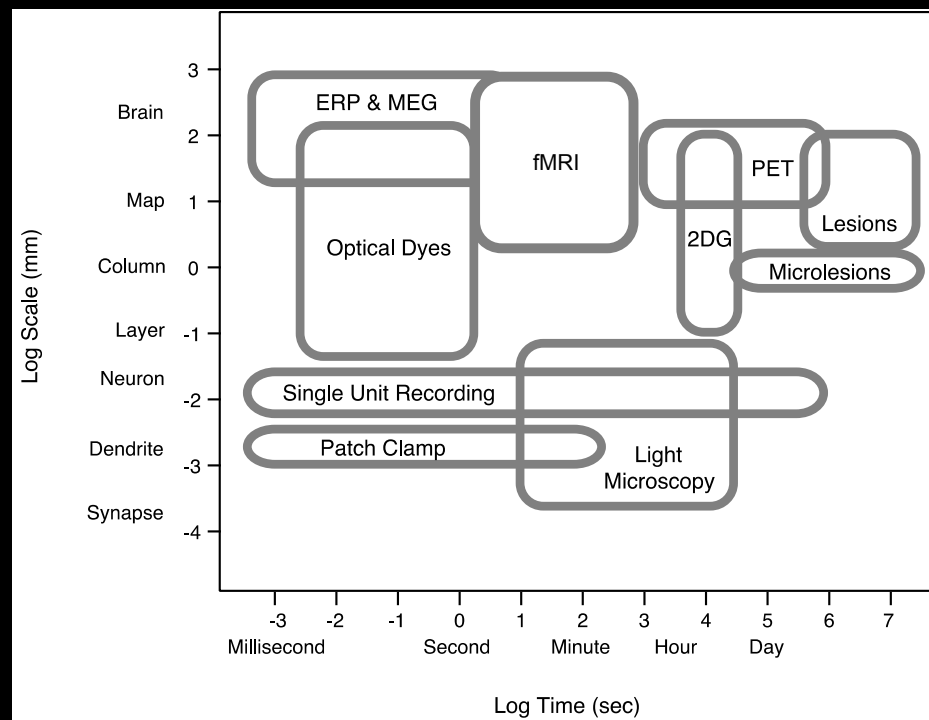
Computation

Functions

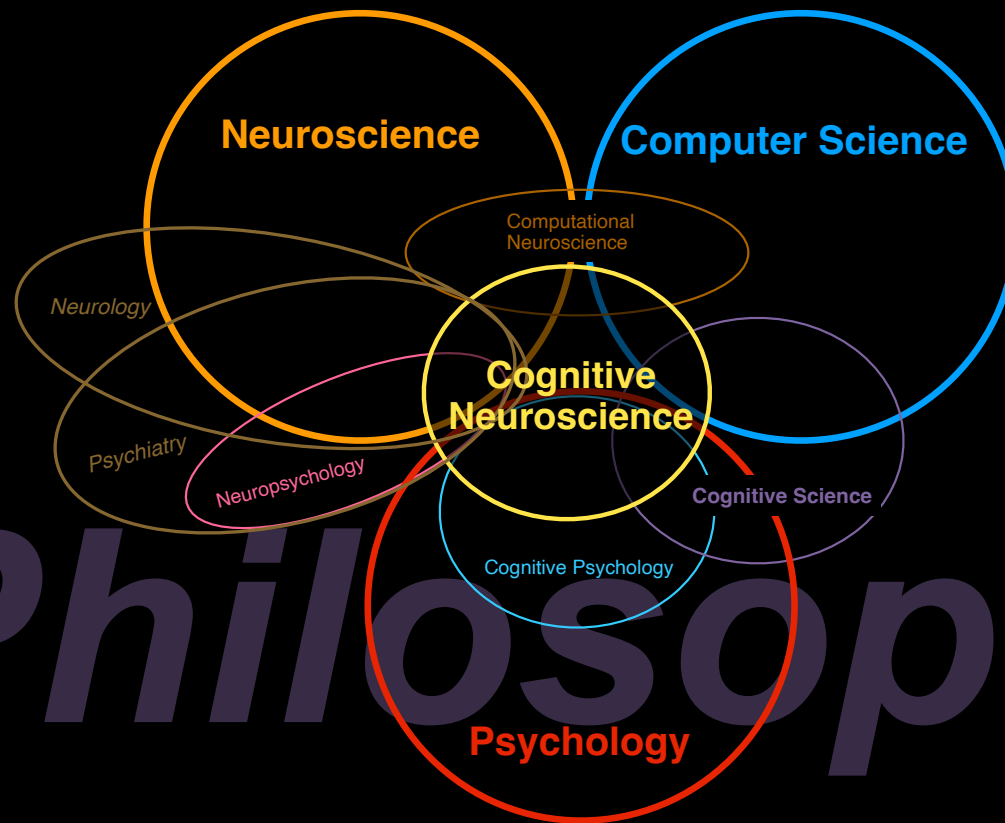
Processes / Representations

Processing units

Levels of Observation



Topography of Fields



Forms of Formalism

Mathematical

Formulas

Solutions

Simple

Precise

Computational

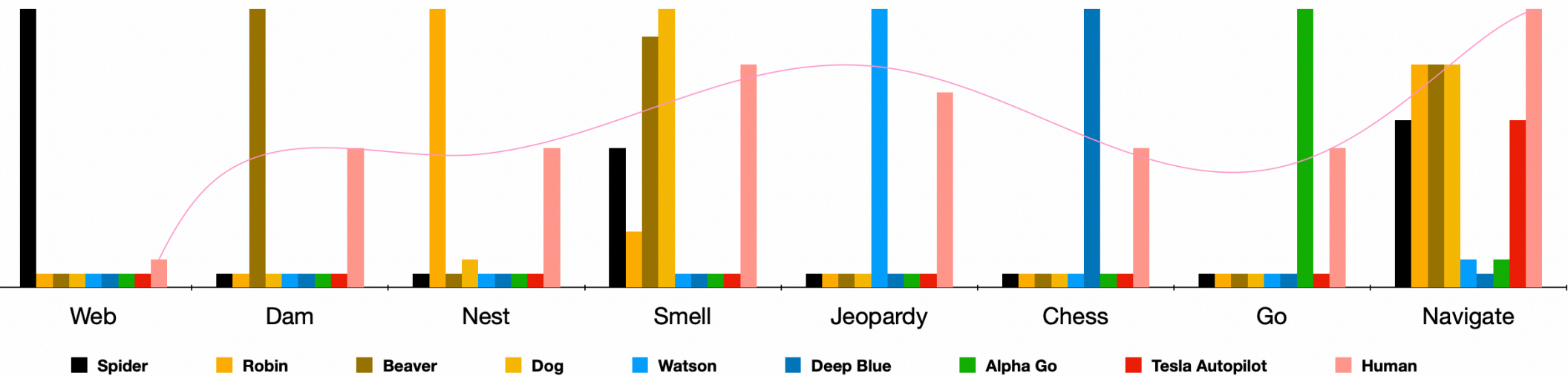
Programs

Simulations

Realistic

Accurate

Contrasts and Tradeoffs



- flexibility vs. efficiency...

“Unexplainable AI”

Turing, 1950

“We also wish to allow the possibility than an engineer or team of engineers may construct a machine which works, but whose manner of operation cannot be satisfactorily described by its constructors because they have applied a method which is largely experimental. “

The Bitter Lesson

Rich Sutton, ~2020

“The second general point to be learned from the bitter lesson is that the actual contents of minds are tremendously, irredeemably complex; we should stop trying to find simple ways to think about the contents of minds, such as simple ways to think about space, objects, multiple agents, or symmetries. All these are part of the arbitrary, intrinsically-complex, outside world. They are not what should be built in, as their complexity is endless; **instead we should build in only the meta-methods that can find and capture this arbitrary complexity.** Essential to these methods is that they can find good approximations, but ***the search for them should be by our methods, not by us. We want AI agents that can discover like we can, not which contain what we have discovered.*** Building in our discoveries only makes it harder to see how the discovering process can be done.”

(<http://www.incompleteideas.net/Incldeas/BitterLesson.html>)